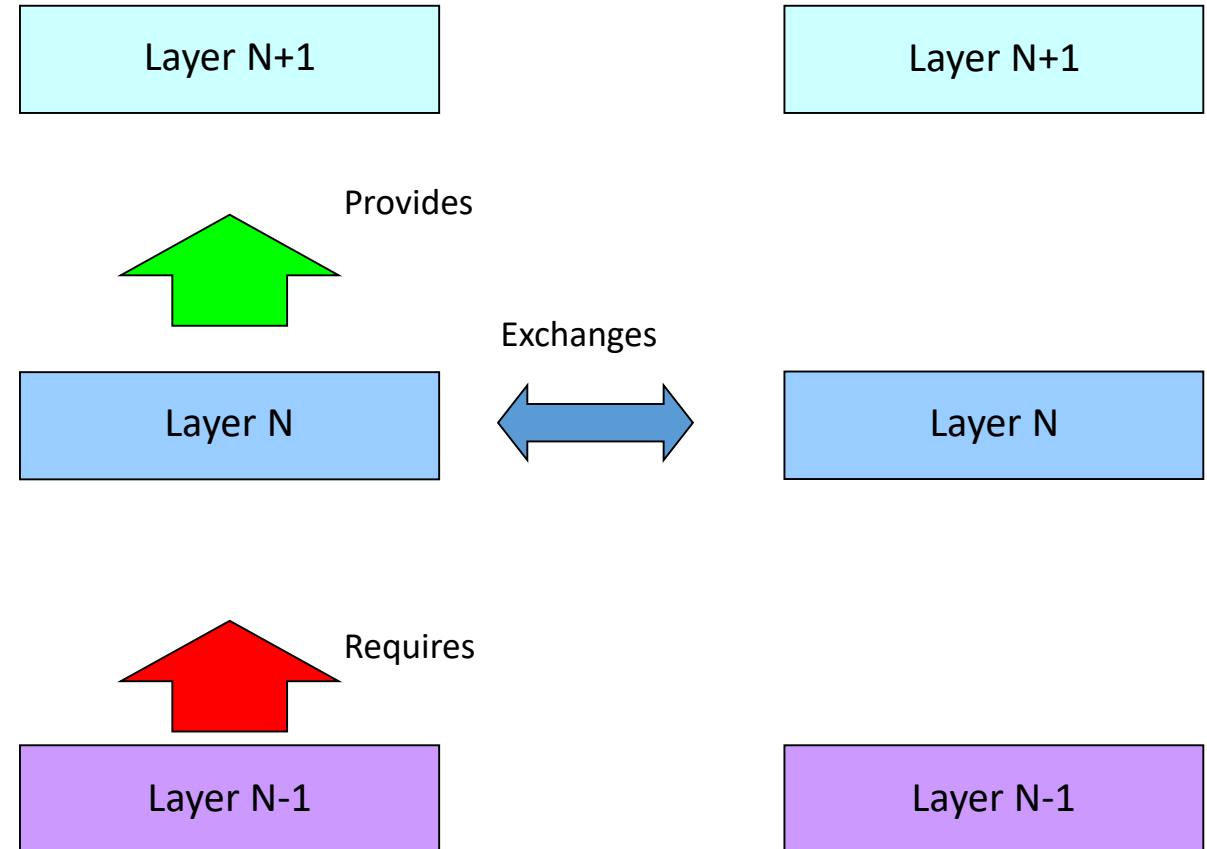




Computer Networks & Communication (BCAC 1011)

Network Models

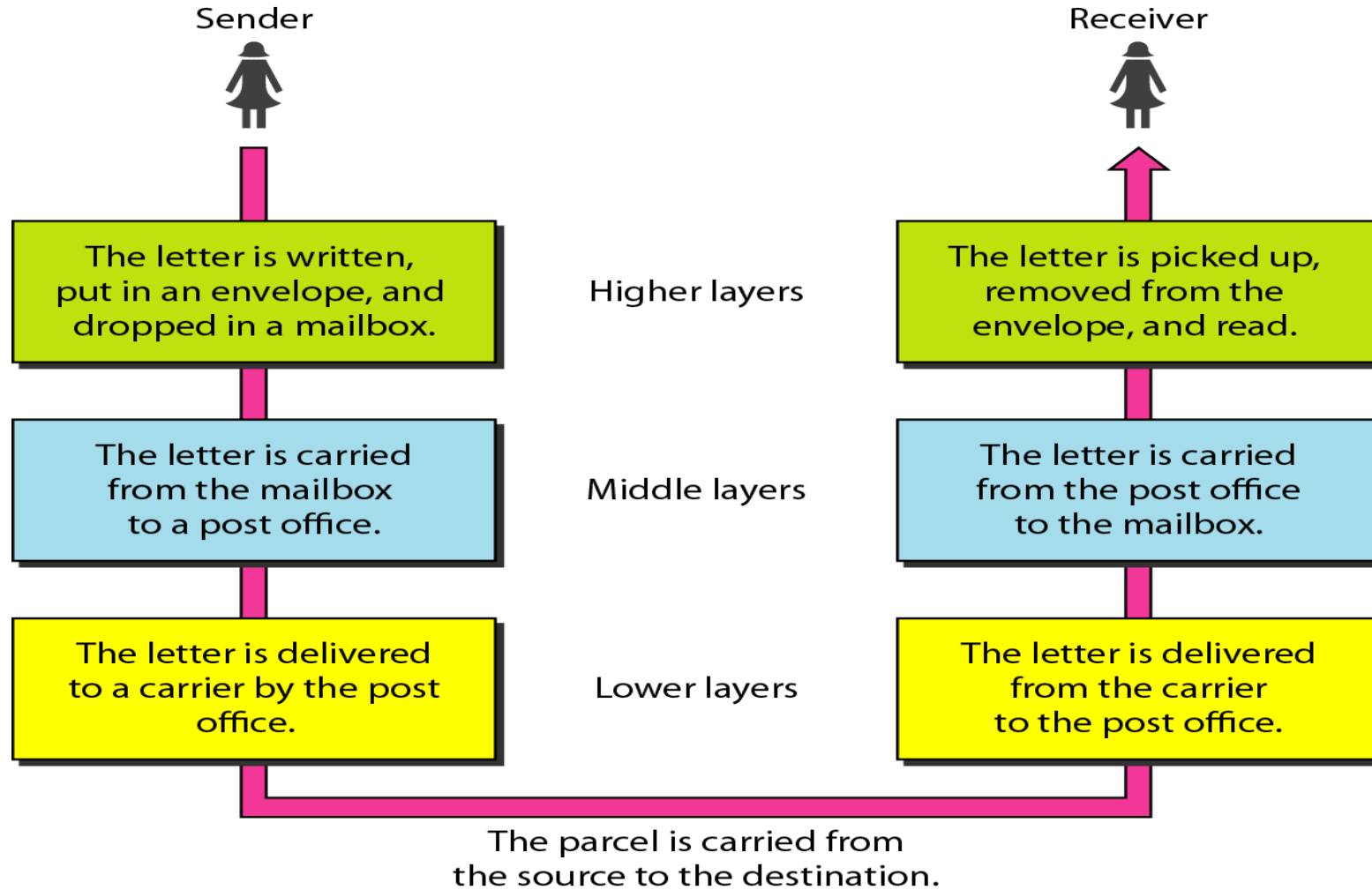
- Network models use **layers** to describe networks
- Each layer describes the **services** provided to the layer above it and those required from the layer below it
- It also describes the format of exchanges between **peer** layers on different network hosts
- Because the layers “**stack**” on top of one another, we often refer to network protocol “stacks” when we talk about the implementation



- **Building complex systems is hard!**
 - Approach: “Divide and conquer”.
 - Split job into smaller jobs, or layers.

Each step dependent on the previous step but does not need to be aware of how the previous step was done.

Tasks involved in sending a letter (Ref Fourouzan)



Task of communication broken up into modules

Network Models

- The most well-known network model is the OSI (Open Systems Interconnect) Reference Model defined and maintained by the Organization for International Standardization (ISO)
- It consists of seven layers, numbered from the bottom (closest the network) to the top (closest the user)

Layer 7 – Application

Layer 6 – Presentation

Layer 5 – Session

Layer 4 – Transport

Layer 3 – Network

Layer 2 – Data Link

Layer 1 – Physical

Physical Layer: Physical layer is the lowest layer of all. It is responsible for sending bits from one computer to another. This layer is not concerned with the meaning of the bits and deals with the physical connection to the network and with transmission. This layer defines electrical and physical details represented as 0 or a 1.

Functions of Physical Layer:

Representation of Bits: Data in this layer consists of stream of bits.

Data Rate: This layer defines the rate of transmission which is the number of bits per second.

Synchronization: It deals with the synchronization of the transmitter and receiver.

Interface: The physical layer defines the transmission interface between devices and transmission medium.

Line Configuration: This layer connects devices with the medium: Point to Point configuration and Multipoint configuration.

Topologies: Devices must be connected using the following topologies: Mesh, Star, Ring and Bus. 14

Transmission Modes: Physical Layer defines the direction of transmission between two devices: Simplex, Half Duplex, Full Duplex.
Deals with baseband and broadband transmission.

Data Layer:

Data link layer is most reliable node to node delivery of data. It forms frames from the packets that are received from network layer and gives it to physical layer.

Error controlling is easily done. Error detection bits are used by the data link layer. It also corrects the errors.

FUNCTIONS OF DATA LINK LAYER:

Framing: Frames are the streams of bits received from the network layer into manageable data units.

Physical Addressing: The Data Link layer adds a header to the frame in order to define physical address of the sender or receiver of the frame, if the frames are to be distributed to different systems on the network.

Flow Control: A flow control mechanism to avoid a fast transmitter from running a slow receiver by buffering the extra bit is provided by flow control. This prevents traffic jam at the receiver side.

Error Control: Error control is achieved by adding a trailer at the end of the frame. Duplication of frames are also prevented by using this mechanism .

Access Control: Protocols of this layer determine which of the devices has control over the link at any given time.

Two Parts of Data Link Layer

Data link layer is divided into two subparts

1. Logical Link Control Layer (LLC)
2. Media Access Control Layer (MAC)

LLC: Logical Link Control layer is the upper layer of the two in data link layer. Essential for the communications between devices. Supports connection oriented and connectionless services both.

MAC: Media Access Control is the lower layer. It is responsible for providing a method for station to gain access to the medium.

The Network Layer :

It routes the signal through different channels from one node to other.

It acts as a **network controller**.

It manages the Subnet traffic.

It decides by which route data should take.

It divides the outgoing messages into **packets**.

Functions of Network Layer:

- It translates logical network address into physical address.
- Routers and gateways operate in the network layer.
- Mechanism is provided by Network Layer for routing the packets to final destination.
- Connection services are provided.
- Breaks larger packets into small packets.

Transport Layer

It decides if data transmission should be on parallel path or single path.

It receives messages from the Session layer above it, convert the message into smaller units and passes it on to the Network layer.

Transport layer breaks the message (data) into small units so that they are handled more efficiently by the network layer.

Function of Transport Layer:

Service Point Addressing : Transport Layer header includes service point address which is port address.

Segmentation and Reassembling : A message is divided into segments; each segment contains sequence number, which enables this layer in reassembling the message.

Flow Control : In this layer, flow control is performed end to end.

Error Control : Error Control is performed end to end in this layer to ensure that the complete message arrives at the receiving transport layer without any error.

Error Correction is done through retransmission.

The Session Layer :

Session layer manages and synchronizes the conversation between two different applications.

Transfer of data from source to destination session layer streams of data are marked and are resynchronized properly, so that the ends of the messages are not cut and data loss is avoided.

Functions of Session Layer:

Dialog Control : This layer allows two systems to start communication with each other in half-duplex or full-duplex.

Synchronization : This layer allows a process to add checkpoints which are considered as synchronization points into stream of data.

Example: If a system is sending a file of 800 pages, adding checkpoints after every 50 pages is recommended. This ensures that 50 page unit is successfully received and acknowledged. This is beneficial at the time of crash as if a crash happens at page number 110; there is no need to retransmit 1 to 100 pages.

The Presentation Layer :

The primary goal of this layer is to take care of the syntax and semantics of the information exchanged between two communicating systems.

Presentation layer takes care that the data is sent in such a way that the receiver will understand the information (data) and will be able to use the data.

Functions of Presentation Layer:

Translation : Before being transmitted, information in the form of characters and numbers should be changed to bit streams.

Encryption : It carries out encryption at the transmitter and decryption at the receiver.

Compression : It carries out data compression to reduce the bandwidth of the data to be transmitted. It is important in transmitting multimedia such as audio, video, text etc.

Application Layer :It is the topmost layer. Transferring of files disturbing the results to the user is also done in this layer. Mail services, directory services, network resource etc are services provided by application layer. This layer mainly holds application programs to act upon the received and to be sent data.

Functions of Application Layer:

Mail Services : This layer provides the basis for E-mail forwarding and storage. ☐

Directory Services : This layer provides access for global information about various services.

File Transfer, Access and Management (FTAM) : It is a standard mechanism to access files and manages it.

Protocols at Application Layer

- **HTTP**:Hyper Text Transfer Protocol
- **FTP**:File Transfer Protocol
- **SMTP**:Simple Mail Transfer Protocol
- **NFS**:Network File Services
- **NVT**:Network Virtual Terminal

Merits of OSI

- OSI model distinguishes well between the services, interfaces and protocols.
- Protocols of OSI model are very well hidden.
- Protocols can be replaced by new protocols as technology changes.
- Supports connection oriented services as well as connectionless service.

Demerits of OSI

- Model was devised before the invention of protocols.
- Fitting of protocols is tedious task.
- It is just used as a reference model.

